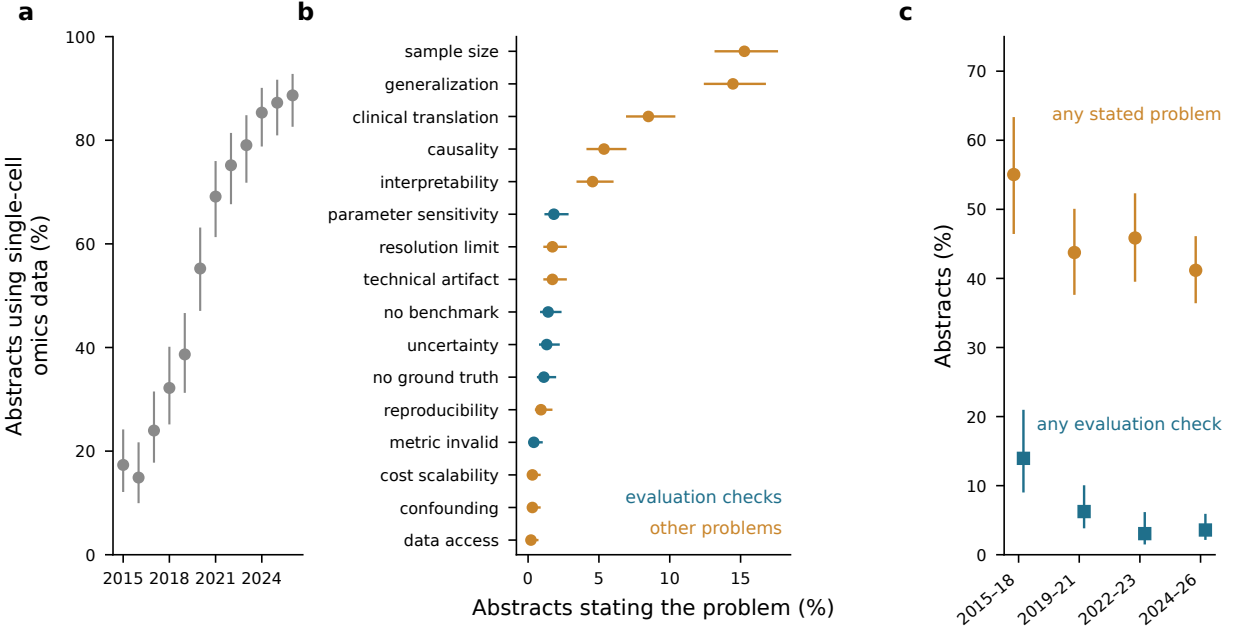
a
b
c
Abstracts using single-cell omics data (%)
100
80
60
40
20
0
2015
2018
2021
2024
sample size
generalization
clinical translation
causality
interpretability
parameter sensitivity
resolution limit
technical artifact
no benchmark
uncertainty
no ground truth
reproducibility
metric invalid
cost scalability
confounding
data access
evaluation checks
other problems
0
5
10
15
Abstracts stating the problem (%)
Abstracts (%)
70
60
50
40
30
20
10
0
any stated problem
any evaluation check
2015–18
2019–21
2022–23
2024–26